low-current combination technique—a major reason why temperature control is possible in LED filament bulbs.

The current scenario

While the market for LED filament bulbs is still developing in India, these bulbs have a growing demand in overseas markets such as Europe. LED filament bulbs come in various shapes (oval, globe and candelabra) and sizes. At present, these are available in warm white and cool white colours.

Making the switch: From Edison to LED

LED filament bulbs offer a variety of benefits to the customers, making them a good choice for replacement. Cost-efficiency is their USP.

Eliminate frequent bulb replacement. One of the many concerns in running a business is keeping the operational costs low. And this includes the cost of the lighting system installed in the store. The heat issue and power output of light have always affected air-conditioning and performance of the bulbs used.

In an incandescent bulb, electricity flowing through the filament generates heat, which turns the filament red. The entire process results in decreasing the life-hours of the bulb. This, in turn, forces the retailer to replace the bulbs more frequently, thus incurring unnecessary costs.

However, LED filament bulbs don't heat up much, which ensures their longevity. In terms of life-hours, an LED filament bulb can light up for a whopping 25,000 hours, while an incandescent bulb lights up for 1000 hours only.

Make it brighter. In an incandescent bulb, only one per cent of the total energy dissipated is converted into light energy, thus resulting in a lower output. In contrast, LED filament bulbs give a higher lumens output as nearly 40 per cent of the energy dissipated by these bulbs is converted into light while the loss of energy is very negligible.

For instance, if a restaurant uses traditional incandescent bulbs in its dining area, at least three to four such bulbs will be needed in just one part of the room in order to keep it bright for customers. However, when using LED filament bulbs, the restaurateur may have to fix only a single bulb in that area as the efficacy of LED filament bulbs is substantially higher than traditional Edison bulbs.

Create the right ambience. Ambience plays a pivotal role in customer service oriented businesses like hospitality and retail stores. Dimmers are crucial in such scenarios. Usually, lifestyle chains, hotels and restaurants have dimmers installed in their stores to create the right ambience for their customers. But since traditional incandescent bulbs are often not compatible with these dimmers, they end up fluctuating frequently, thus rendering dimmers completely useless. Nevertheless, lighting majors have addressed this issue in LED filament bulbs, making them compatible with most of the dimmers.

To understand, consider the situation in which a customer wants to have a romantic dinner with his partner at your restaurant but is put off by the 100-watt traditional incandescent lights everywhere. While you would want to serve this customer, unfortunately, you do not have the right tools to fulfil this need. Ultimately, you end up losing a customer because of your 100-watt Edison bulbs. Now, imagine dimming the lights with LED filament bulbs to create the right ambience. You have not only won a new customer but also probably a returning customer and a great word-of-mouth promotion for your restaurant.

Longer durability. We explained above how LED filament bulbs have a lower maintenance cost than typical incandescent bulbs. Another contributing factor is that LED filament bulbs are far more durable than their incandescent counterparts: The tungsten filament of incandescent bulbs is very fragile and can break even with very little mishandling of the bulb.

The glitch: high initial cost

While LED filament bulbs use a high-end technology and offer several benefits to customers, unfortunately, these are exorbitantly priced and can burn a hole in buyers' pockets. On an average, an LED filament bulb costs nearly 100 per cent more than an incandescent bulb and at least 20 per cent more than the ordinary LED bulb.

Justifying the high price of the product, Kumar explains that since most lighting players do not have the requisite infrastructure to manufacture the components required for LED filament bulbs, they have to import the components from abroad.

"Initially this product is going to be costly as we have signed agreements with LED suppliers and took permission for patents regarding the gas mixture to be used in glass bulb. Later, the price of the product will come down as the need for heat sinks for heat dissipation has been eliminated completely, resulting in a lightweight lamp that costs less," he shares.

At present, LED light manufacturers source LED filaments mainly from China.

Are LED filament bulbs here to stay?

Industry experts suggest that the market for LED filament bulbs will grow rapidly as customers are increasingly looking at long-term benefits rather than just the initial cost. According to a market report by Epistar, the LED filament bulb market is expected to reach up to 300 million bulbs by 2017 and double to 600 million bulbs by 2018. And if the numbers are too hard to believe, wait and watch your competitors lap up rave reviews for their sudden aestheticism while you struggle to get a few more customers! EFY